Numerical simulation of flow around a two dimensional square cylinder through $\kappa - \omega$ turbulence model

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Abstract

This study investigates the behavior of the standard κ–ω turbulence model from a physical and numerical perspective. A simulation of flow around a two-dimensional square cylinder is carried out using the finite volume method in ANSYS Fluent. The outcome is compared with other results and experimental data found in literature, but also with a previous simulation obtained with the exact same setup except for the turbulence model, the κ–ε RNG. This final comparison will show that κ–ω can capture some of the main flow features but it is not so precise and complete as the κ–ε RNG and the other methods of the state of the art.

1. Introduction

The accurate prediction of turbulent flows around bluff bodies, such as square cylinders, is a critical challenge in various engineering disciplines. However, the inherent complexity and multi-scale nature of turbulence necessitate the use of mathematical models to take into account the effects of these unpredictable fluctuations in the Reynolds-Averaged Navier-Stokes equations. Among these, two-equation models like the κ-ε and κ-ω models are commonly employed. This study delves into the behavior of the standard κ-ω model from both a physical and numerical perspective, specifically in the context of flow around a two-dimensional square cylinder. The primary objectives are to investigate the capability of the standard κ-ω turbulence model in capturing key flow phenomena, comparing the simulation results with established experimental data and numerical benchmarks from the existing literature, but also conduct a direct comparison with a prior simulation, referred to as DISMA, which utilized the exact same numerical setup but employed the κ-ε RNG turbulence model, which is a widely used alternative.

The remainder of this report is structured as follows: Section 2 provides a detailed mathematical description of the standard κ-ω turbulence model and outlines the numerical methods and settings employed in the simulation. Section 3 presents and analyzes the results, including integral parameters, pressure coefficients, and velocity profiles. A comprehensive comparison with the DISMA simulation and literature data is presented in Subsections 3.1 and 3.2, respectively.

2. Turbolence model and settings

The turbolence model used in this study is the standard two equation $\kappa - \omega$ one. It is a RANS (Reynolds Averaged Navier Stokes) approach where the turbolence fluttuations are neglected in order to drastically reduce the computational cost of the numerical simulation. The RANS equation is founded by applying the Reynolds decomposition (a generic variable ϕ is decomposed as $\phi = \bar{\phi} + \phi'$ that is the sum of a averaged and a fluctuating component) to the variables of the Navier-Stokes equation.

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_i}\left[\nu\left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i}\right)\right] - \frac{\partial}{\partial x_j}(\overline{u_i' u_j'}), \qquad \frac{\partial \bar{u}_i}{\partial x_i} = 0$$

In the RANS equations it is possible to find the Navier Stokes for averaged variables and a new term which refers to the fluctuating component. This new term is named Reynolds tensor and it is defined as $R_{ij} = \overline{u_i' u_j'}$. Note that the new obtained system is not closed and it is necessary to introduce some semi-empirical equation to fix this issue. Turbolence model are introduced in order to find some strategies to approximate the Reynolds tensor effect on average flux following a physical intuition. In this case, it is proposed the standard $\kappa - \omega$ model which tries to approximate the tensor using the Boussinesq hypotesis:

$$\overline{u_i' u_j'} = \nu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i}\right)$$

where ν_t is called eddy viscosity. After this assumption, the Reynolds tensor is approximated to a scalar, so a 3D anisotropous flux is considered isotropous. The physical meaning of this variable is also unclear, and it depends on the flux rather than on the fluid properties such as the kinematic viscosity ν. Analyzing the orders of magnitude, it is established the following relation for the eddy viscosity.

$$\nu_t \propto \Lambda \cdot u_\Lambda \quad \left[\frac{m^2}{s}\right]$$

It is necessary to have 2 equation in order to find the new variables.

- $u_\Lambda \propto \kappa^{\frac{1}{2}} \left[\frac{m}{s}\right]$ where κ is the turbolent kinetic energy.
- Λ should be founded considering the dissipation ε or, as it is presented in this paper, with the specific dissipation rate ω.

Considering the specific dissipation as $\omega \propto \dfrac{\varepsilon}{\kappa} = [s^{-1}]$, the viscosity of the eddy is now expressed as $\nu_t = \dfrac{\kappa}{\omega}$. Actually, by experimental results it is known that the proportionality constant is $\beta_* = 0.09$ such that $\omega = \dfrac{1}{\beta^*}\dfrac{\varepsilon}{\kappa}$. Moreover, it is needed one equation for κ and an other for ω. For the turbolent kinetic energy it is derived from the Navier Stokes equation with these simple operations.

From RANS equation it is subtracted the classic NS equation, after that it is multiplied by u_i' and in the end the time average is computed. The equation for ω si constructed in analogy to κ with some extra term in order to preserve dimensionality.

$$\begin{cases} \dfrac{\partial \kappa}{\partial t} + \dfrac{\partial \bar{u}_i \kappa}{\partial x_j} = \dfrac{\partial}{\partial x_j}\left[\left(\dfrac{\nu_t}{\sigma_\kappa^\omega} + \nu\right)\dfrac{\partial \kappa}{\partial x_j}\right] + P_\kappa - \underbrace{\beta^* \omega \kappa}_{\varepsilon} \\ \dfrac{\partial \omega}{\partial t} + \dfrac{\partial \bar{u}_j \omega}{\partial x_j} = \dfrac{\partial}{\partial x_j}\left[\left(\dfrac{\nu_t}{\sigma_\omega} + \nu\right)\dfrac{\partial \omega}{\partial x_j}\right] + \underbrace{\dfrac{\omega}{\kappa} C_{\omega_1} P_\kappa}_{P_\omega} - \underbrace{C_{\omega_2}\omega^2}_{\varepsilon_\omega} \end{cases} \tag{1}$$

The values of the semi-empirical constants are:

$$\beta^* = 0.09, \qquad C_{\omega_1} = \frac{5}{9}, \qquad C_{\omega_2} = \frac{3}{40}, \qquad \sigma_\kappa^\omega = \sigma_\omega = 2.$$

While all the terms that needed to be modeled after the time average that leads to system 1 are the following:

- The production of κ through $P_\kappa = -\overline{u_i' u_j'}\dfrac{\partial \bar{u}_i}{\partial x_j}$, that is modeled using Boussinesq hypotesis again:

  $$-\overline{u_i' u_j'}\frac{\partial \bar{u}_i}{\partial x_j} \approx \nu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i}\right)\frac{\partial \bar{u}_i}{\partial x_j}$$

  It is always positive because $sign\left(\dfrac{\partial \bar{u}}{\partial y}\right) \neq sign(\overline{uv})$. P_κ can be seen as a term which extract energy form the averaged flux.

- The diffusion term D_κ is composed by

  $$\frac{\partial}{\partial x_j}\left[\frac{\overline{p' u_j'}}{\rho} + \frac{1}{2}\overline{u_j' u_i' u_i'} + \nu \frac{\partial \kappa}{\partial x_j}\right] \approx \frac{\partial}{\partial x_j}\left[\left(\frac{\nu_t}{\sigma_\kappa^\omega} + \nu\right)\frac{\partial \kappa}{\partial x_j}\right].$$

  The pressure term is neglected and, using the gradient hypotesis, κ is transported from high κ zones to low κ zones, so $\dfrac{1}{2}\overline{u_j' u_i' u_i'} = \dfrac{1}{2}\overline{u_j' \kappa} \approx \dfrac{\nu_t}{\sigma_\kappa^\omega}\dfrac{\partial \kappa}{\partial x_j}$.

- In the end, $\beta^* \omega \kappa$ is the dissipative term and ω has his transport equation.

In the same way of κ the ω equation is formalized. Compared to the $\kappa - \varepsilon$, in the stagnation zone the fluctuating components of the velocity, and consequently the kinetic energy κ, tend to zero. So P_k is near to zero, while P_ω, and ε are κ-indipendent. Hence correctly the eddy viscosity ν_t tends to zero. This chain of consequences, derived from the introduction of the equation for ω instead of ε, avoid the problem of the overproduction of kinetic energy near the stagnation region, which is a crucial effect of the $\kappa - \varepsilon$ standard model applied to bluff bodies.

2.1. Numerical methods and settings

In this subsection it is resumed all the other parameters are used in our simulation. The Reynolds number is fixed at 21400 and it is considered incompressible flow impacting the cylinder with zero angle of attack. To numerically solve the problem Ansys Fluent is used, adopting the finite-volume method. The numeric scheme used in the spacial discretization is QUICK for each equation, while the algorithm used to solve the system is the PISO one. The time discretization follows the second order Implicit Euler scheme. About the boundary, there are non-slip conditions on the longer edges and on the body surface and the flux goes from left to right entering in the computational domain with one unite of velocity (the simulation is performed in a non dimensional setting). A Dirchlet inlet boundary condition is imposed following the semi-empirical results in [1] such that $It = 2\%$. So $\kappa = \frac{3}{2}(\bar{U}It)^2 = 0.0006$ and, assuming the turbulence integral length scale $L_T = 0.5B$, $\omega \approx \frac{\sqrt{\kappa}}{L_T} \approx 0.049$. The mesh used to discretize the domain is hybrid, there is a small rectangular area of $15B \times 13B$ around the square which is made of an unstructured triangular grid, while the bigger rectangular area that covers the rest of the domain is made of a structured quadrangular gird. The total lenght is $45B$ ($15B$ on the left of the square and 29 on the right) and the total height is $30B$ ($14.5B$ above and $14.5B$ Below the square).

3. Results

The simulation approximately reach convergence after $t = 150$ that, using a time step $\Delta t = 0.02$, is equivalent to 7500 iterations.

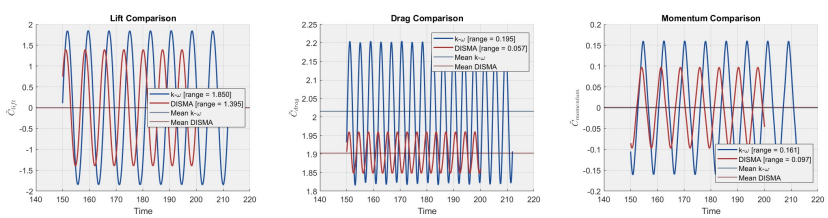

Figure 1: Lift (left), Drag (center) and Momentum (right) after convergence.

The period and the Strouhal number of vortex shedding are $T = 7.76$, $St = \frac{1}{T}\frac{B}{U} = 0.129$. The lift coefficients C_Y, drag coefficient C_X and momentum coefficient C_M are reported in tab.1. The mean and RSM of the pressure coefficient C_p is also calculated (fig 3). The mean of the speed $\bar{u}_x$ along $\frac{y}{B} = 0$ is presented.

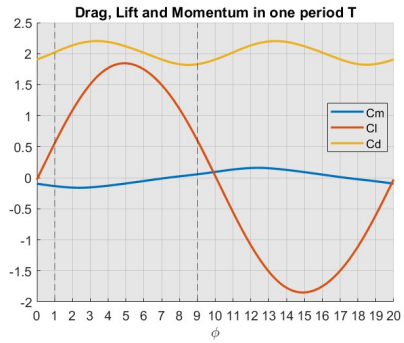

Figure 2: Lift, Drag and Momentum for one period T

Authors	St	$\bar{C}_X$	$\tilde{C}_X$	$\tilde{C}_Y$
Lyn	0.135	2.05-2.23	n.a.	n.a.
Sakamoto	0.134	2.22	0.132	0.145
Durao	0.139	n.a.	n.a.	n.a.
Murakami	0.132	2.09	0.13	1.60
DISMA	0.138	1.90	0.057	1.40
$\kappa - \omega$ [2]	0.131	1.96	n.a.	n.a.
$\kappa - \omega$ (SST) [2]	0.126	2.13	n.a.	n.a.
$\kappa - \omega$ (our)	0.129	2.02	0.195	1.85

Table 1: Comparison of Integral values.

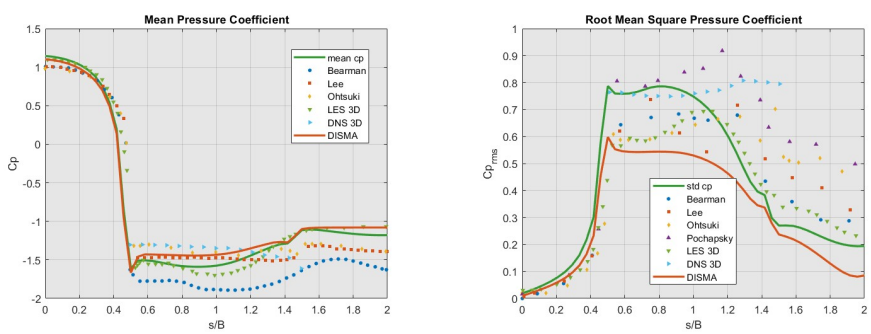

Figure 3: Mean and RSM of C_p.

Furthermore, a comparison between the instantaneous values of the variables is presented at two of the 20 phases (ϕ_1, ϕ_9) where the Lyn measurement was performed. In particular, the instantaneous value of u_y along the line $\frac{y}{B} = 0$ and the instantaneous value of u_x along the line $\frac{x}{B} = 1$ are shown.

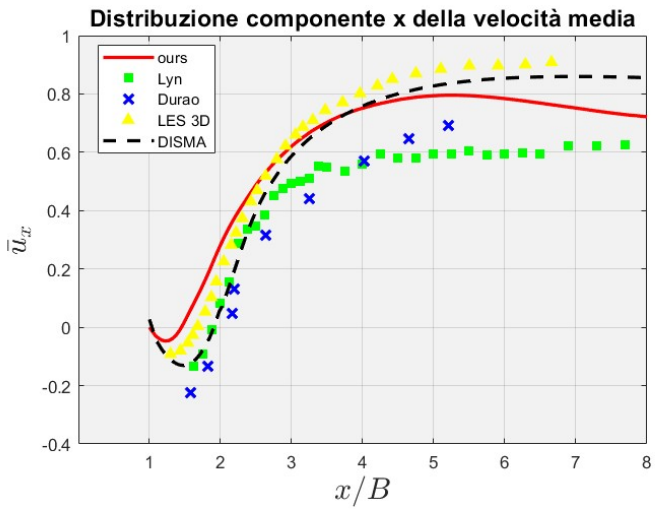

Figure 4: $\bar{u}_x$ over $\frac{y}{B} = 0$

3.1. Comparison with DISMA simulation

In this subsection the $\kappa - \varepsilon$ RNG DISMA simulation results are compared with the one presented in this study. It is important to say that the DISMA numerical settings are the same as the one adopted in this study. So the difference between the two results can be related only to the turbolence model. Starting from the integral variables (Table 1), note that the Strouhal number is lower than the DISMA one which is similar to the Lyn experiment. On the other hand, $\kappa - \omega$ seems to be more accurate on the value of $\bar{C}_X$. There is a big difference between the oscillation amplitude $\tilde{C}_X, \tilde{C}_Y$. Also the RSM of C_p is very different, while its mean value is quite similar (Fig 3). Analyzing the mean $\bar{u}_x$ over $\frac{y}{B} = 0$ (Fig 4) it is clear that $\kappa - \omega$ cannot describe so well the streamwise velocity in the recirculation zone like the DISMA does, underestimating it. After this region, our values get closer to the others, but then, after $\frac{x}{B} = 5$, they follow a different path, decreasing and converging near 0.6.

However, considering the instantaneous value of u_y over $\frac{y}{B} = 0$ (Fig 5) and u_x over $\frac{x}{B} = 1$ (Fig 6) it is possible to see that the DISMA and our $\kappa - \omega$ simulation are very close. The only difference is in the intensity of u_y near the cylinder for phase one, which is slightly overestimated in our model with respect to DISMA simulation. Connecting this observation to the previous one regarding the underestimation of $\bar{u}_x$ in the same region, it seams that the kinetic energy is almost the same between the two models. It seams just differently distributed among the two components of the velocity.

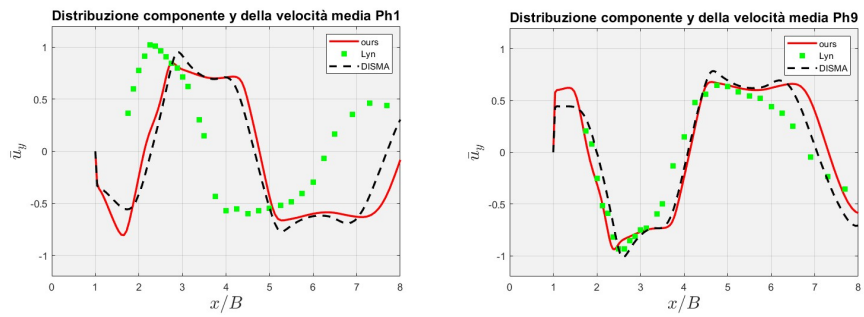

Figure 5: u_y over $\frac{y}{B} = 0$ for ϕ_1, ϕ_9

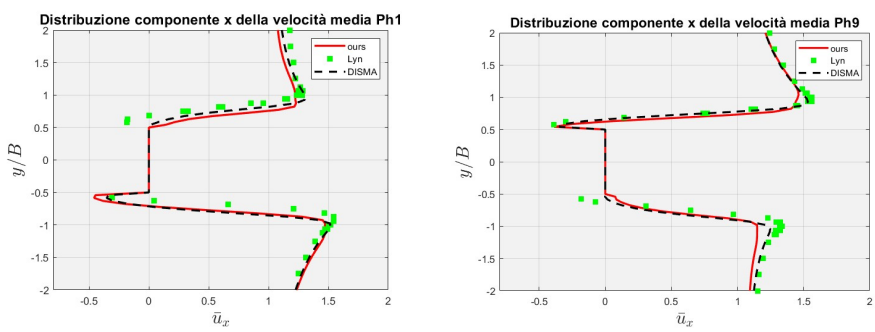

Figure 6: u_x over $\frac{x}{B} = 1$ for ϕ_1, ϕ_9

3.2. Comparison with literature

Here we want to compare the results of our simulation with the referance values found in literature. Starting from the distribution of the longitudinal component of the mean velocity $\bar{u}_x$ along the midline of the square, can be seen that our model underestimate the lenght of the recirculation region (Fig.4) but also the intensity of the velocity inside that. However, after this region, our collection is quite consistent with the scatter of literature data, being right in the middle between LES results and Lyn measurement from $x/B = 3$ on. Nonetheless, it is the only one which do not have a monotonic convergence to the wake velocity far from the body. With reference to figure 5, that shows the distribution of $\bar{u}_y$ along the same midline, the difference from Lyn is significant. Our model overestimate the lenght of the coordinate in which the velocity is maximum, but also the one where it gets to the minimum. It produces a region between $x/B = 3.5$ and $x/B = 4.5$ in which the velocity gets to a plateau, slightly increase and then rapidly decrease. While, according to Lyn, velocity goes to a maximum and than monotonically decreases. Something specular happen in the minimum point. This is true both for phase one and nine. The reason for that could be the lack of enough kinetic dissipation in the wake, which leads to this higher intensity of velocity. Moreover, leading the attention to the

$\bar{u}_x$ just behind the square (Fig.6), can be noticed that there is recirculation at the height of the boundary layer only on one side of the square at a time (between phase one and nine) while Lyn data show reverse flow on both of the sides simultaneously. This points out that our model fails to capture the three-dimensional nature of the turbulent fluctuations, which are instead observed in experiments and persist across the various phases. All the incorrect prediction over the flow, are reflected in the fluctuations of the integral parameters, in particular the drag and lift coefficients, which are quite overestimated compared to the literature. The C_y gets to 1.85 but the highest value among the state of the art is just 1.60; and even if the mean of C_X is quite in line with other data, its rsm is $\approx 50\%$ bigger. On the other hand, the Strouhal is almost the same of the literature results.

4. Conclusion

In this report, the κ-ω turbulence model was used to simulate the flow around a square cylinder and compared with both literature data and DISMA results. Our simulation provides a good prediction of the mean drag coefficient and Strouhal number, aligning well with several literature values. It successfully captures the fundamental characteristics of the wake flow, such as the periodic vortex shedding. However, the model exhibits certain limitations. It overestimates force fluctuations and underestimates recirculation length, struggling to accurately represent the intensity of velocities and the wake recovery. The direct comparison with the DISMA simulation suggests that the κ-ε RNG model, under similar numerical conditions, might offer a more complete and precise description of certain flow features, such as the dynamic within the wake. Overall, the standard κ-ω model provides reasonable predictions for some global parameters but its accuracy is limited in complex regions. For improved precision in such applications, maintaining a comparable computational cost, more advanced turbulence models are necessary, such as the κ-ε RNG. Being aware that the selection of the turbulence model depends on the specific flow phenomena and the desired level of accuracy. Further investigation could be conducted on more sophisticated variation of κ-ω, such as the SST. Either, there could be something else in the numerical setting that can reduce the errors of the standard one if modified ad hoc. Many of these possible combination, and their mutual effects on the results, are still to be explored.

References

[1] D. Lyn, W. Rodi, The flapping shear layer formed by flow separation from the forward corner of a square cylinder, Journal of Fluid Mechanics 267 (1994) 353–376. doi:10.1017/S0022112094001173.

[2] A. Ying, B. Xu, C. Zhang, Numerical simulation of three-dimensional flow around a square cylinder at reynolds numbers up to $10^5, Computers\&Fluids 44(2012)276 - -289. \; doi : 10.1016/j.compfluid.2011.10.012$.